VENTURE SNAPSHOT

HaLow Rescue

Infrastructure-Free Emergency UAV Communication for Flood and Disaster Zones

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Competition

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Mays Business School, Texas A&M University

100M+ people displaced by floods annually	**72 hrs** critical rescue window	**5** drone mesh network	**32.5 Mbps** peak throughput no infrastructure	**0** infra-free solutions with range + speed

01.

Problem and Opportunity

When floods and landslides strike, communication infrastructure fails first. Cell towers go down, roads are blocked, and rescue teams lose coordination at the exact moment lives depend on it. Over 100 million people are displaced by floods annually, and the critical window for survival is 72 hours.

No existing infrastructure-free technology delivers both the range and throughput that emergency UAV coordination requires. LoRa reaches 5 km but transmits less than 0.01 Mbps. Standard RC links offer 1.5 km at 0.1 Mbps. 5G delivers 100 Mbps but requires the very infrastructure that floods destroy. The gap is real, documented, and unaddressed.

HaLow Rescue fills that gap with a five-drone Wi-Fi HaLow 802.11ah mesh network delivering 32.5 Mbps at up to 1 km with zero infrastructure required. A Random Forest machine learning layer trained on RSSI, SNR, and MCS index telemetry predicts link degradation before failure occurs, enabling proactive drone repositioning. The full dataset, trained model, and Python scripts will be published open-source on GitHub under Creative Commons, multiplying global impact beyond any single deployment.

02.

Starting Point Snapshot

Venture Stage: Pre-revenue. Hardware partially built.

- Two MM6108-EKH05 Morse Micro Wi-Fi HaLow evaluation boards acquired.
- Drone frame design underway using 3D-printed PETG at 250 to 300mm wheelbase.

- Faculty advisor confirmed: Dr. Marcelo Menezes De Carvalho, Texas State University Ingram School of Engineering.
- Target field dataset: 700 to 1,100 labeled UAV link measurements across multiple distances and obstruction conditions.

03.

Highest-Risk Assumptions

1 **Wi-Fi HaLow sustains reliable throughput in real outdoor UAV conditions**

The 32.5 Mbps specification comes from controlled datasheet conditions. Real-world flight introduces multipath interference, altitude variation, and weather effects. This assumption must be stress-tested in the field before any rescue deployment claim can be made.

2 **A Random Forest classifier trained on lab telemetry generalizes to unseen disaster environments**

The ML model will be trained on data collected at Texas State. Flood zones in Nepal, Bangladesh, or the Philippines present different terrain, humidity, and obstruction profiles. Generalization is not guaranteed and must be validated independently.

3 **Emergency response organizations will adopt a student-built, open-source UAV system**

Technical capability alone does not drive adoption. First responder agencies carry procurement processes, liability concerns, and training requirements. Whether HaLow Rescue integrates into existing emergency response workflows is currently unproven.

4 **The five-drone mesh topology maintains network integrity when individual nodes fail**

Autonomous mesh rerouting under real flight conditions has not been tested. Node failure during a rescue operation is a high-stakes scenario the current architecture has not been validated against.

5 **Open-source release drives replication and impact at scale**

Publishing to GitHub is a values-driven decision. Whether other teams actually download, replicate, and deploy the system requires active community engagement that has not yet been established.

04.

Experimentation Roadmap

Phase 1 Exploration (Current)	Phase 2 Ground Testing Fall 2026	Phase 3 Field Development Fall 2026 > Spring 2027	Phase 4 Validation Spring 2027	Phase 5 Release Spring 2027
Problem scoped. Hardware acquired. Advisor engaged. Mesh topology designed.	All five nodes bench-tested. Firmware validated. Baseline data collected.	Outdoor flight tests. 700+ labeled link measurements. ML model trained.	Mesh rerouting tested under node failure. Formation-hold validated. F1 documented.	Open-source GitHub publish. IEEE conference submission. Community outreach.

Teal phases are active or in progress. Grey phases are planned for Spring 2027.

05.

Planned Use of AI to Accelerate Learning

At the product level

A Random Forest classifier runs on-board a Raspberry Pi 4 and processes real-time RSSI, SNR, and MCS index telemetry from each drone node. When the model predicts link degradation above a defined confidence threshold, it triggers a repositioning alert before the link fails. This shifts the system from reactive to proactive, which in a 72-hour rescue window is the difference that matters.

At the venture learning level

Claude (Anthropic) was used as a thought partner throughout the design process to pressure-test assumptions, identify gaps in the risk framework, and stress-test the logic of the experimentation roadmap. All strategic decisions remained with the founding team. AI was used to sharpen thinking, not to replace it.

At the impact level

The open-source release is itself an AI acceleration strategy. By publishing the dataset and model freely, HaLow Rescue enables teams globally to fine-tune the classifier on their own regional terrain data, creating a distributed learning network that improves the model far faster than any single team could achieve alone.

06.

Platform Expansion Potential

The current build establishes the core mesh communication and ML prediction layer. Once validated, the same platform is designed to expand in three natural directions without requiring a fundamental redesign.

Live video over HaLow

The Morse Micro MM6108-EKH05 evaluation board includes an onboard camera interface. Future builds can stream live video from each drone node back to the ground station over the existing HaLow link, giving rescue coordinators real-time visual situational awareness with no additional communication hardware required.

Larger mesh networks

The five-drone topology is the proof-of-concept floor, not the ceiling. The 802.11ah standard supports mesh scaling. Additional nodes can be added to cover larger disaster zones, with the ML layer adapting to route around degraded links across a wider network.

Integration with existing emergency systems

The open dataset and standardized telemetry format are designed to be compatible with existing UAV coordination platforms used by first responder agencies, lowering the barrier to real-world adoption.
